Reg. No. | | | | | | | | | | | [14EE661]

SIXTH SEMESTER B. Tech EXAMINATION, JUNE–2019

OBJECT ORIENTED PROGRAMMING USING C++

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions choosing atleast **TWO** full questions from each part
2. Missing data may be suitably assumed.

PART-A

1.	a.	Explain the difference between a pointer and a reference variable with a program	08
	b.	List the advantages of using object oriented programming over procedure oriented languages.	06
2.	a.	Write a C++ program to illustrate the passing objects to functions and returning objects from functions.	07
	b.	With the help of syntax and general representation of a Class, explain the terms class, access specifiers, data members and member functions.	07
3.	a.	Define constructor. List the various types of constructor in C++ and write a C++ program illustrate the usage of a copy constructor.	08
	b.	Explain the syntax of a generic function. Write a program to swap two integers and characters using a generic function.	06
4.	a.	What are the pitfalls of operator overloading? List the operators that cannot be overloaded in C++.	06
	b.	Outline the concept of mutable and explicit and write a program to make the constructor explicit with the help of explicit keyword.	08

PART-B

5.	a.	With an example demonstrate how an array can be used as a class member with input and output operations on the array member.	08
	b.	Illustrate the function that can be used from the C++ string class to find the location of a string object.	06
6.	a.	Define inheritance. With a neat diagram of an example and a program, illustrate multiple and multilevel inheritance.	09
	b.	Write a C++ program to illustrate the member functions of base class can be accessed by the objects of the derived class.	05

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

7.	**a.**	Define virtual functions. Explain how run time polymorphism is implemented by using virtual functions with an example program.	**07**
	b.	Describe the applications of pure virtual functions with an example.	**07**
8.	**a.**	Demonstrate a simple exception handling using built in keywords with an example program	**06**
	b.	List and explain the usage of vectors and their advantages over arrays.	**08**